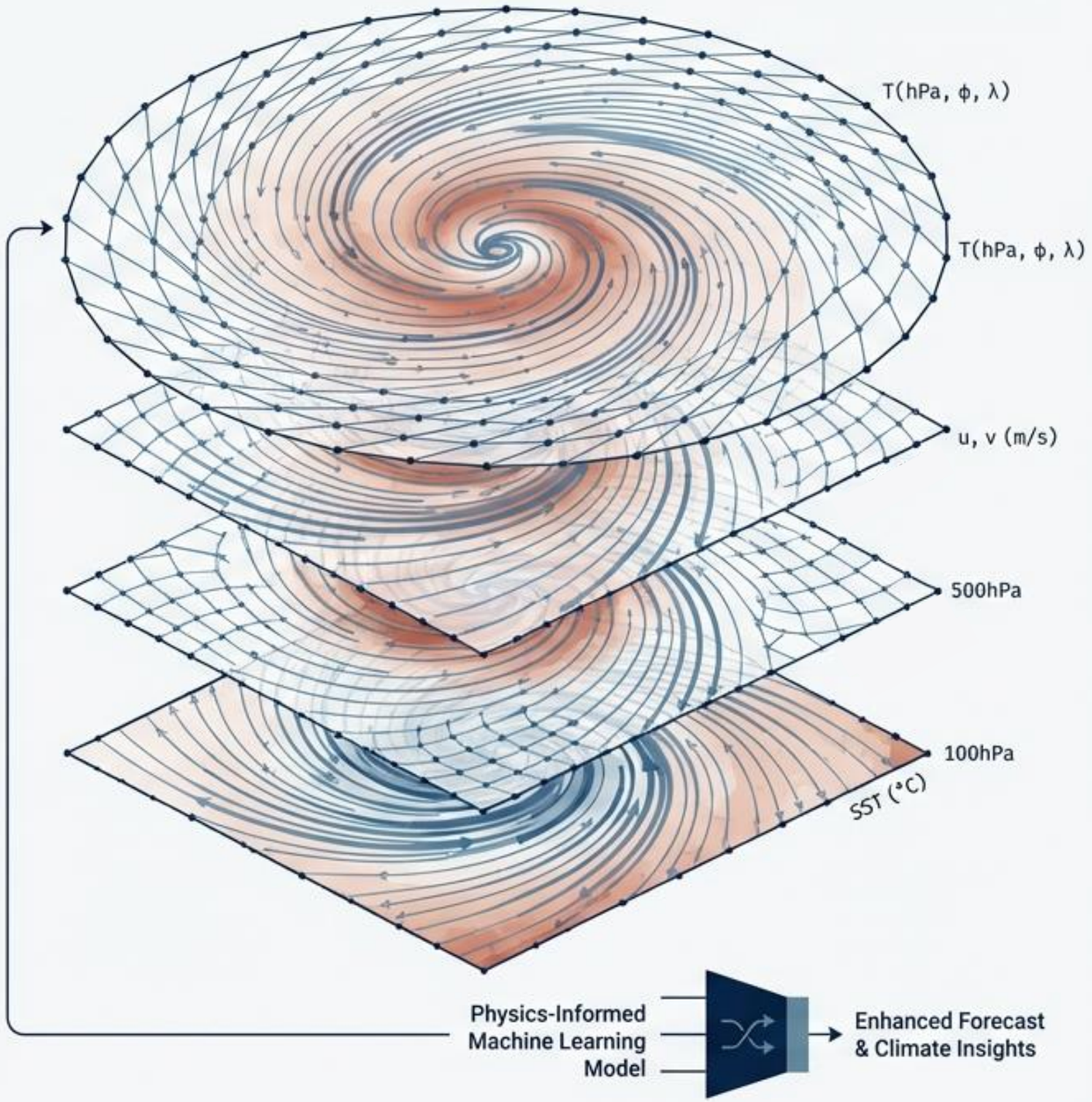
Predicting Tropical Cyclones: An Introduction to TropiCycloneNet

Integrating complex atmospheric dynamics with advanced computational approaches for enhanced forecasting accuracy.

Hadrian Fung



The Thermodynamic Fuel of Climate Change

Rising sea surface temperatures act as thermodynamic fuel, drastically increasing the frequency of catastrophic Category 4 and 5 storms. [1]

[9]

$51 Billion

Average annual direct economic losses globally.

[6]

20.4 Million

People affected annually by extreme cyclone events.

[6]

1.5°C

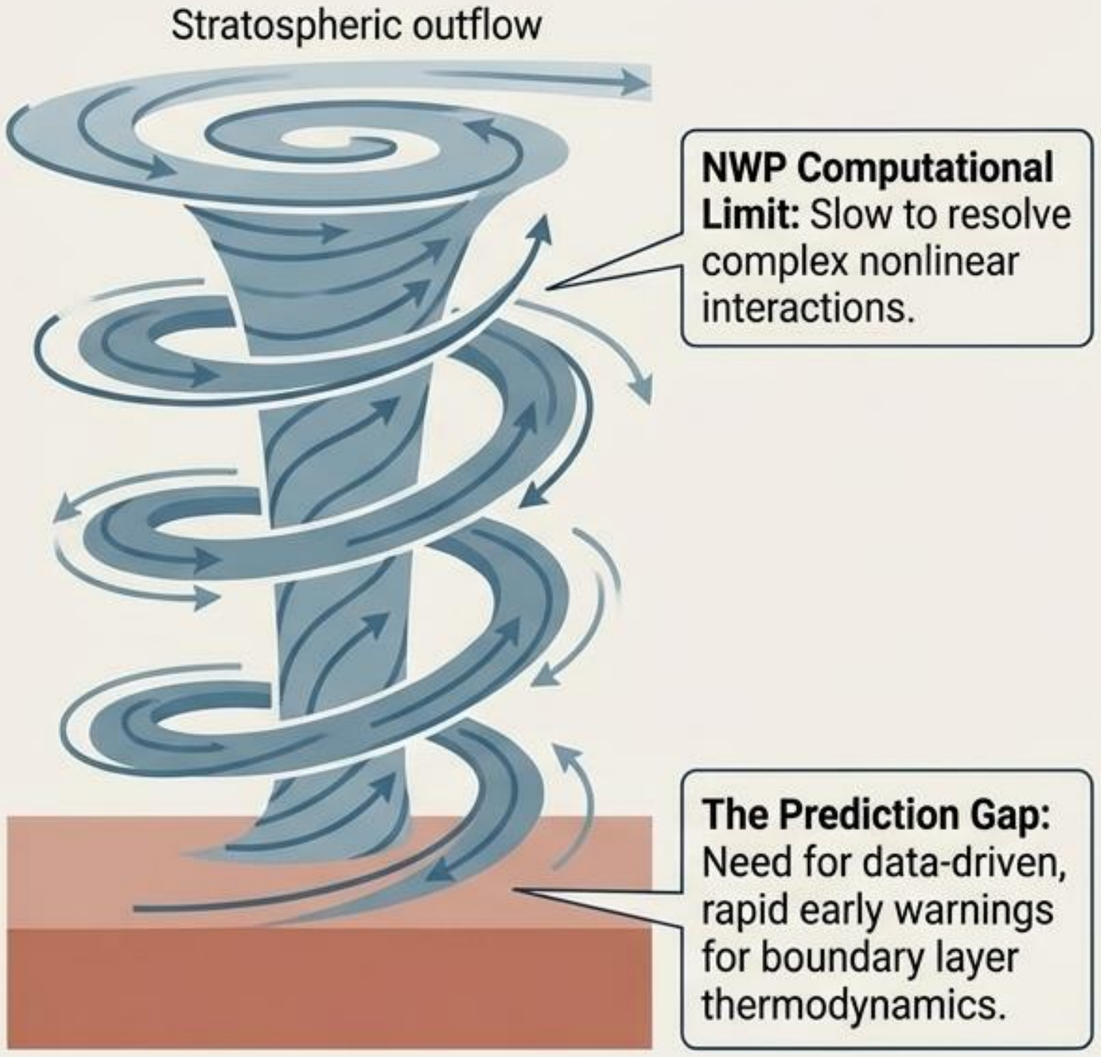
2024 was the warmest year in a 175-year observational history, accelerating rapid intensification.

[1]

The Prediction Gap in Intensity Forecasting

Traditional Numerical Weather Prediction (NWP) models rely on computationally heavy partial differential equations. While track forecasting has improved, intensity forecasting—specifically Rapid Intensification (RI)—severely lags.

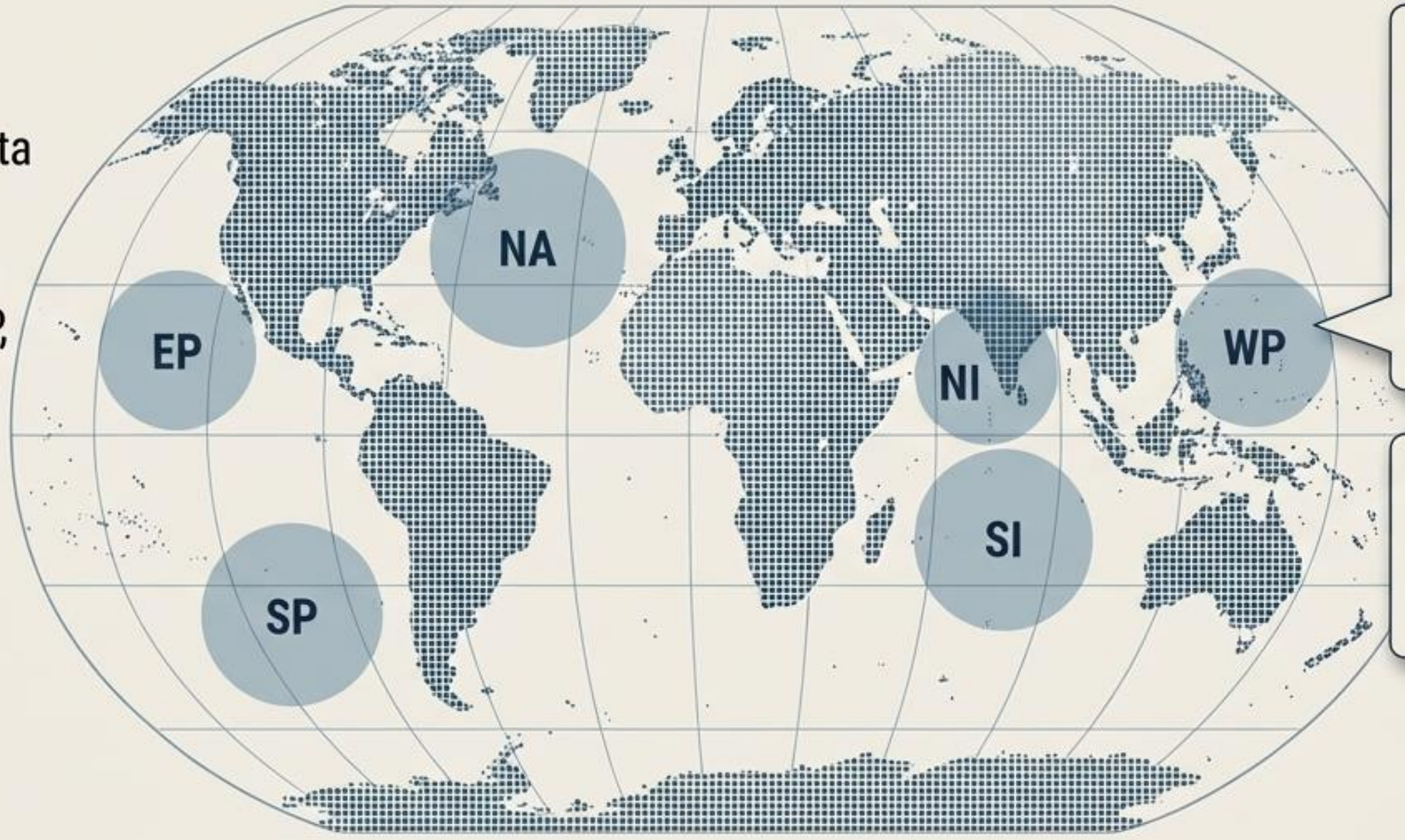
NWPs struggle to resolve the mesoscale (small-scale) thermodynamic processes at the ocean-atmosphere boundary in time for early warnings.



Introducing the TropiCycloneNet Dataset (TCND)

Scale:

- 70 years of data (1950–2023)
- 6 major ocean basins (WP, EP, NA, NI, SI, SP)



Basin Index:

NA: North Atlantic
EP: Eastern Pacific
WP: Western Pacific
NI: North Indian
SI: South Indian
SP: South Pacific

The Test Subset

- 3.34 GB targeted slice (2017–2023)

The Multimodal Dataset

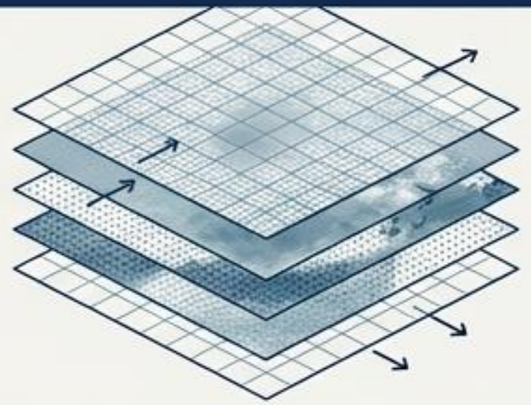
TCND is structured into **three distinct pillars**, allowing modular integration of tabular, spatial, and environmental features.

Data_1d

Location	Pressure	Wind	...
...	...	77.7	...
...	...	152.3	...
...	...	35.5	...
...	...	26.6	...
...	...	...	...
...	...	...	...
...	...	...	...
...	...	...	...

Data_1d: Ground truth tabular records from IBTrACS (Location, Pressure, Wind).

Data_3d



Data_3d: 20° x 20° gridded spatial patches at 0.25° resolution in 6-hour intervals.

Env-Data

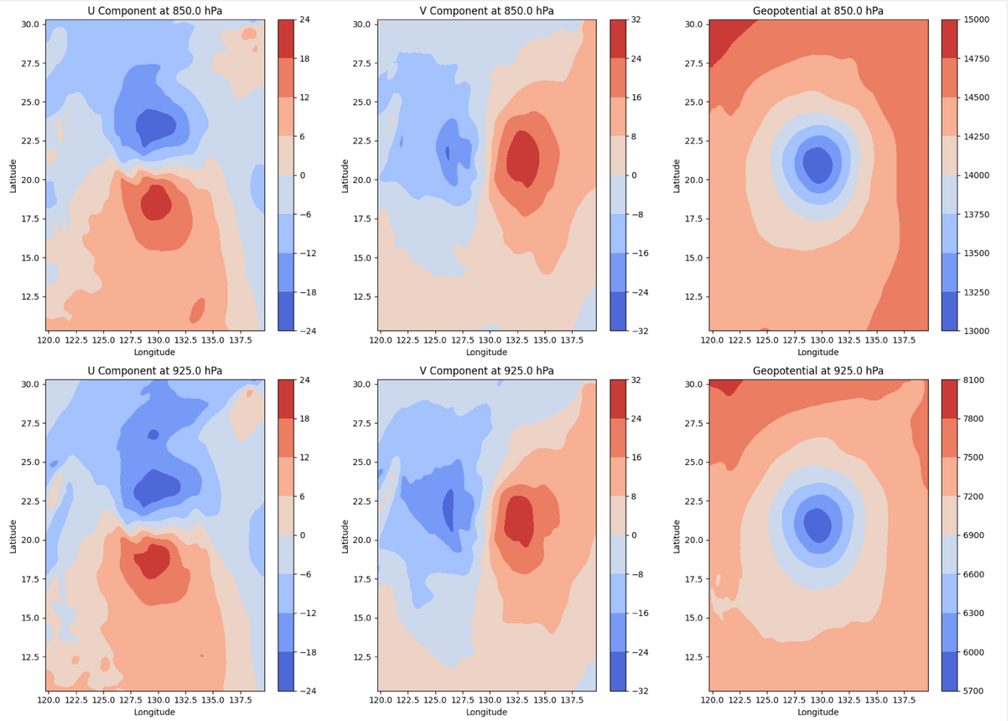
Env-Data: Pre-calculated physical shortcuts and historical momentum.

Unified ML Input

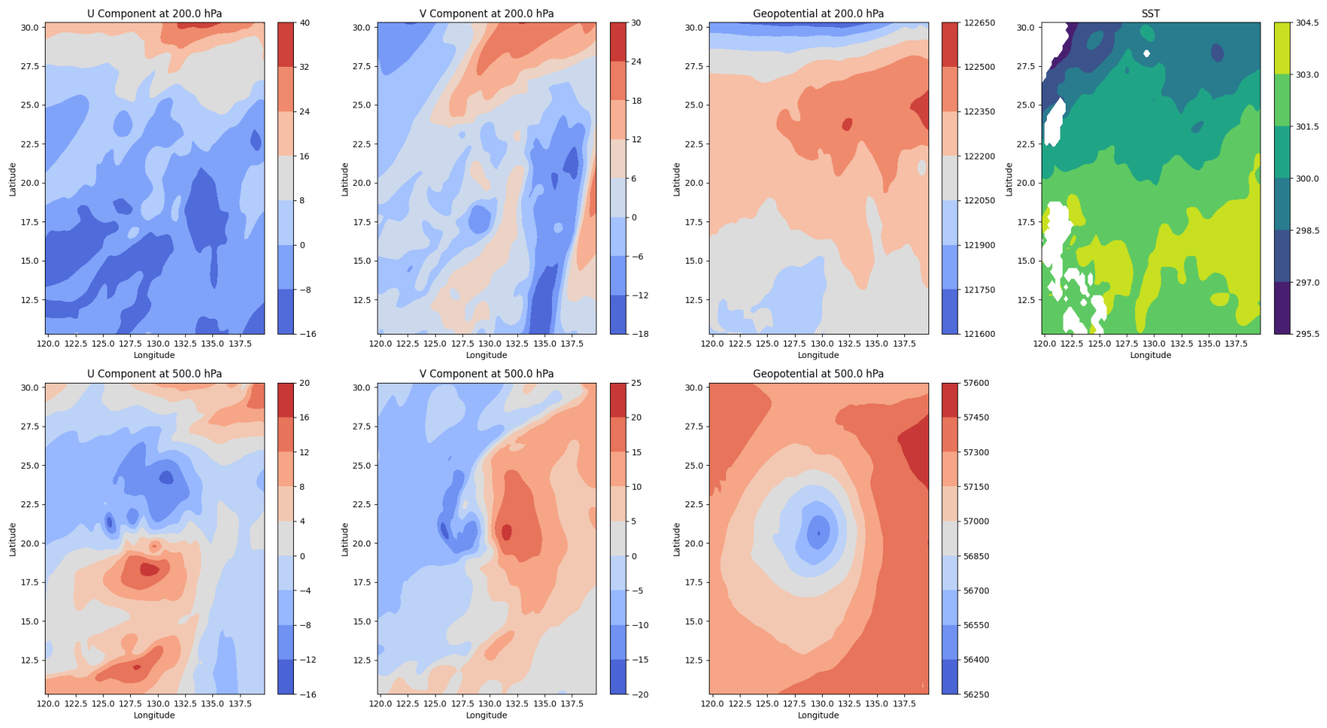
Data availability (1): $Data_{3d}$

Data Name	Sea surface temperature	Geopotential height	U-component of wind	V-component of wind
Spatial resolution	0.25°	0.25°	0.25°	0.25°
Time resolution	6 hourly	6 hourly	6 hourly	6 hourly
Pressure Level (hPa)	/	200, 500, 850, 925	200, 500, 850, 925	200, 500, 850, 925
Size (around the TC center)	81*81 (20°*20°)	81*81 (20°*20°)	81*81 (20°*20°)	81*81 (20°*20°)

[3] [11]



[3]



[3]

Data availability (2): Data_{1d}

- **ID**: Time step of the TC.
- **LONG**: Longitude of the TC centre (with a precision of **0.1°E**).
- **LAT**: Latitude of the TC centre (with a precision of **0.1°N**).
- **PRES**: Minimum pressure (hPa) near the TC centre.
- **WND**: Two-minute mean maximum sustained wind (MSW; m/s) near the TC centre.
- **YYYYMMDDHH**: Date and time (UTC) of the TC event.
- **Name**: Name of the TC.

ID	LONG	LAT	PRES	WND	YYYYMMDDHH	Name
1.0	**-10.28**	**4.36**	**0.3**	**-0.28**	**2001101418**	**Haiyan**
1.0	-10.48	4.46	0.3	-0.28	2001101500	Haiyan
1.0	-10.7	4.54	0.2	-0.2	2001101506	Haiyan
1.0	-10.84	4.62	0.2	-0.2	2001101512	Haiyan
1.0	-10.98	4.78	0.1	0.0	2001101518	Haiyan
1.0	-11.12	4.9	0.1	0.0	2001101600	Haiyan

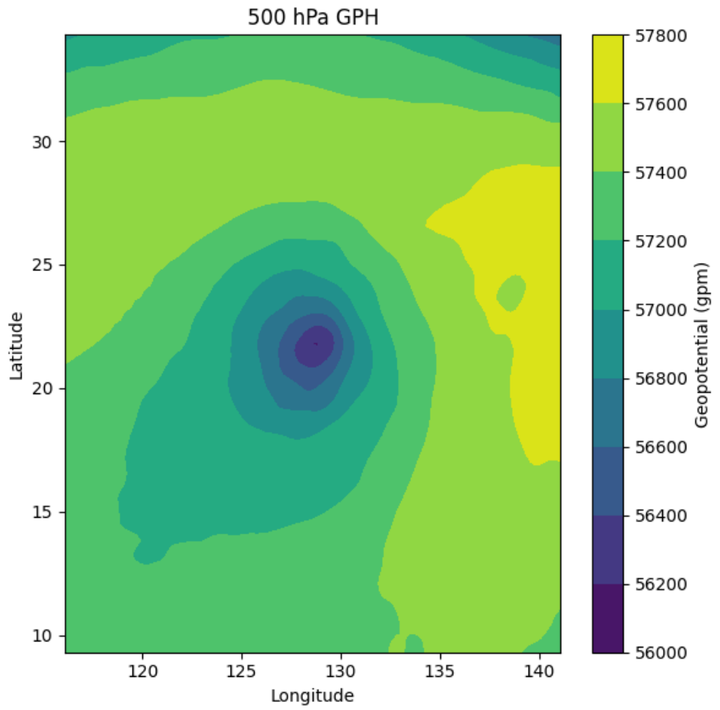
[3]

Data availability (3): Env-Data

- **Movement Velocity**: The movement velocity of the tropical cyclone.
- **Month**: Month of occurrence.
- **Location Longitude and Latitude**: The relative location on Earth.
- **24-hour History of Direction**: The movement direction of the cyclone in the past 24 hours.
- **24-hour History of Intensity Change**: The intensity change of the cyclone in the past 24 hours.
- **Subtropical High Region**: Extracted from **500 hPa Geopotential Height (GPH)** data.

  This variable is processed to make the data more suitable for input to **DL models**.

Key	Value
Moving Velocity	0.09389602136004332
Month	[0. 0. 0. 0. 0. 0. 0. 0. 0. 1. 0. 0.]
Location Longitude	[0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 1. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
Location Latitude	[0. 0. 0. 0. 0. 0. 0. 0. 1. 0. 0. 0.]
History Direction (24 h)	[0. 0. 0. 0. 0. 1. 0. 0.]
History Intensity Change (24 h)	[1. 0. 0. 0.]
Subtropical High	GPH



[3]

Data availability (3): Env-Data

```
{
 'area': array([0., 0., 0., 0., 0., 1.]),
 'wind': 0.4090909090909091,
 'intensity_class': array([0., 0., 0., 0., 1., 0.]),
 'move_velocity': 0.1350808404696287,
 'month': array([0., 0., 0., 0., 0., 0., 0., 0., 1., 0., 0., 0.]),
 'location': (1443.0, 141.0),
 'location_long': array([0., 0., ... 1., ... 0.]),
 'location_lat': array([0., 0., ... 1., ... 0.]),
 'history_direction12': array([0., 0., 0., 0., 1., 0., 0., 0.]),
 'history_direction24': array([0., 0., 0., 0., 1., 0., 0., 0.]),
 'history_inte_change24': array([1., 0., 0., 0.]),

 # Hidden Target Labels
 'future_direction24': 4,
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```

1. Cyclone Attributes & Time

- `month`: A **One-Hot Encoded** array of shape `(12,)` indicating the month of occurrence. *(e.g. `[0,0,0,0,0,0,0,0,1,0,0,0]` means September).*
- `area`: A **One-Hot Encoded** array of shape `(6,)` grouping which of the 6 ocean regions the cyclone belongs to. *(e.g. `[0,0,0,0,0,1]` might map to Western Pacific 'WP').*
- `intensity_class`: A **One-Hot Encoded** array `(6,)` binning the tropical cyclone's intensity category at this exact timestamp (from Tropical Depression up to Cat 5).

2. Locational Positioning

- `location`: The exact Longitude and Latitude values normalized as `(lat * 10, lon * 10)`. *(e.g. `(1443.0, 141.0)` represents roughly `144.3°E`, `14.1°N`).*
- `location_long` / `location_lat`: The authors have quantized the Earth's grid and provided Longitude and Latitude as **One-Hot Encoded blocks** to make them easier for Dense/Embedding network architectures to learn from (shape `(36,)` and `(12,)` respectively).

3. Historical Movement & Kinetics

- `wind`: Normalized intensity metric representing the 2-minute average maximum sustained wind (MSW). *(e.g. `0.409`).*
- `move_velocity`: The normalized speed at which the entire cyclone storm system is moving territorially across the map.
- `history_direction12` & `history_direction24`: **One-Hot Encoded arrays** of shape `(8,)` essentially acting as a compass. They bin the direction the storm has been moving over the past 12 and 24 hours into 8 cardinal direction sectors (N, NE, E, SE, S, SW, W, NW).
- `history_inte_change24`: A **One-Hot array** of shape `(4,)`. It categorizes how rapidly the storm's intensity has changed in the preceding 24 hours (e.g. rapid intensification vs weakening).

4. Ground-Truth (Supervised labels inside the file)

- `future_direction24`: An integer indicating the ground truth direction classification (out of 8 directions) the storm will go in the *next* 24 hours. (Used as the actual target `Y` when training track predictions).
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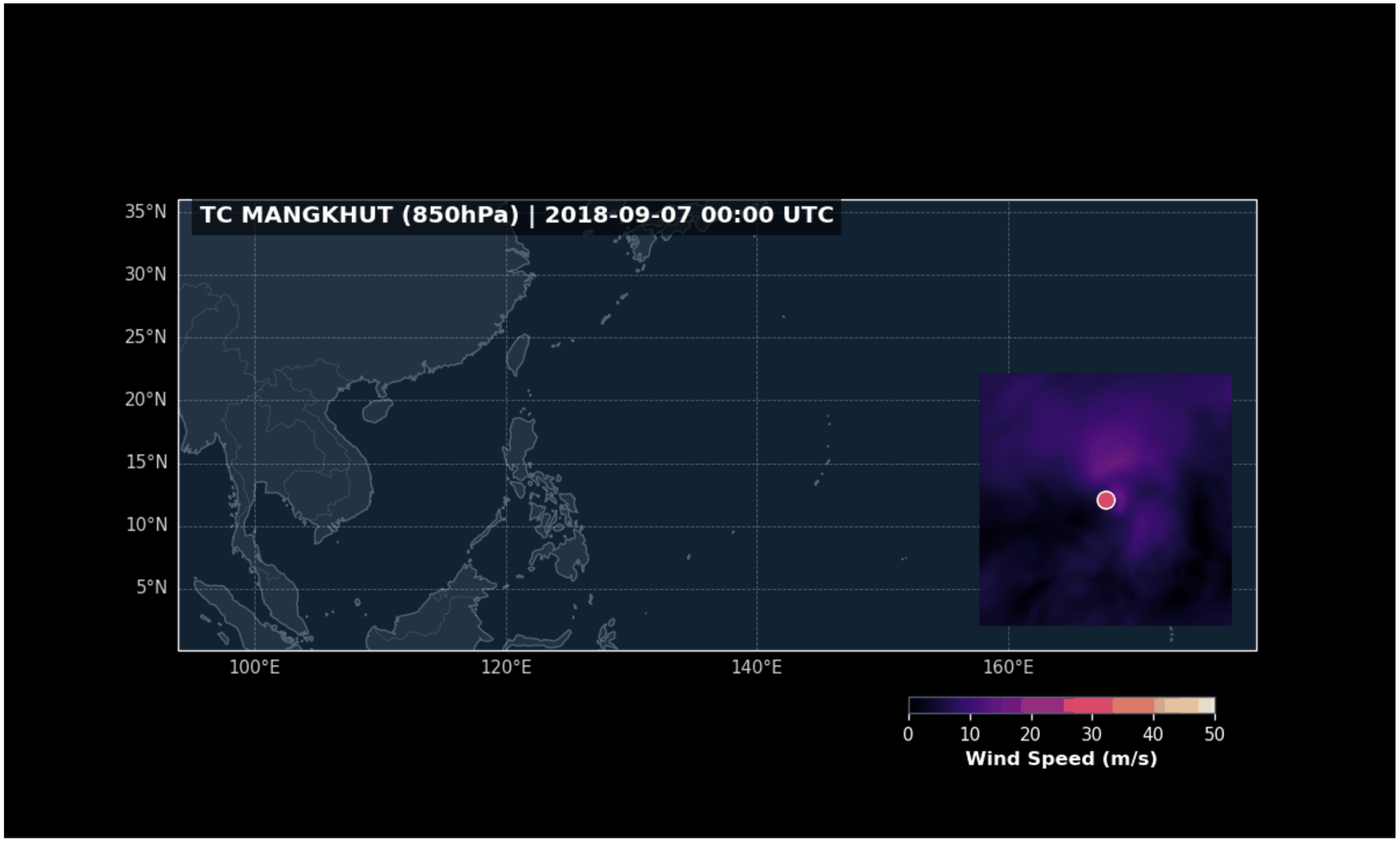
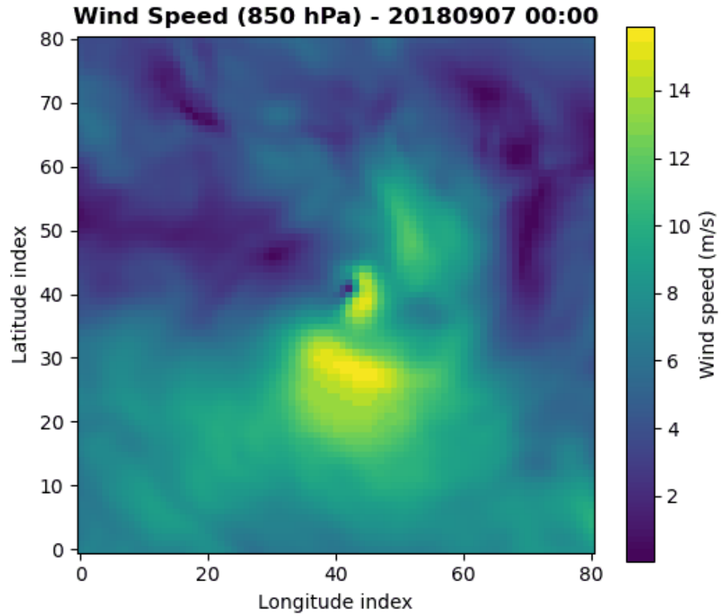
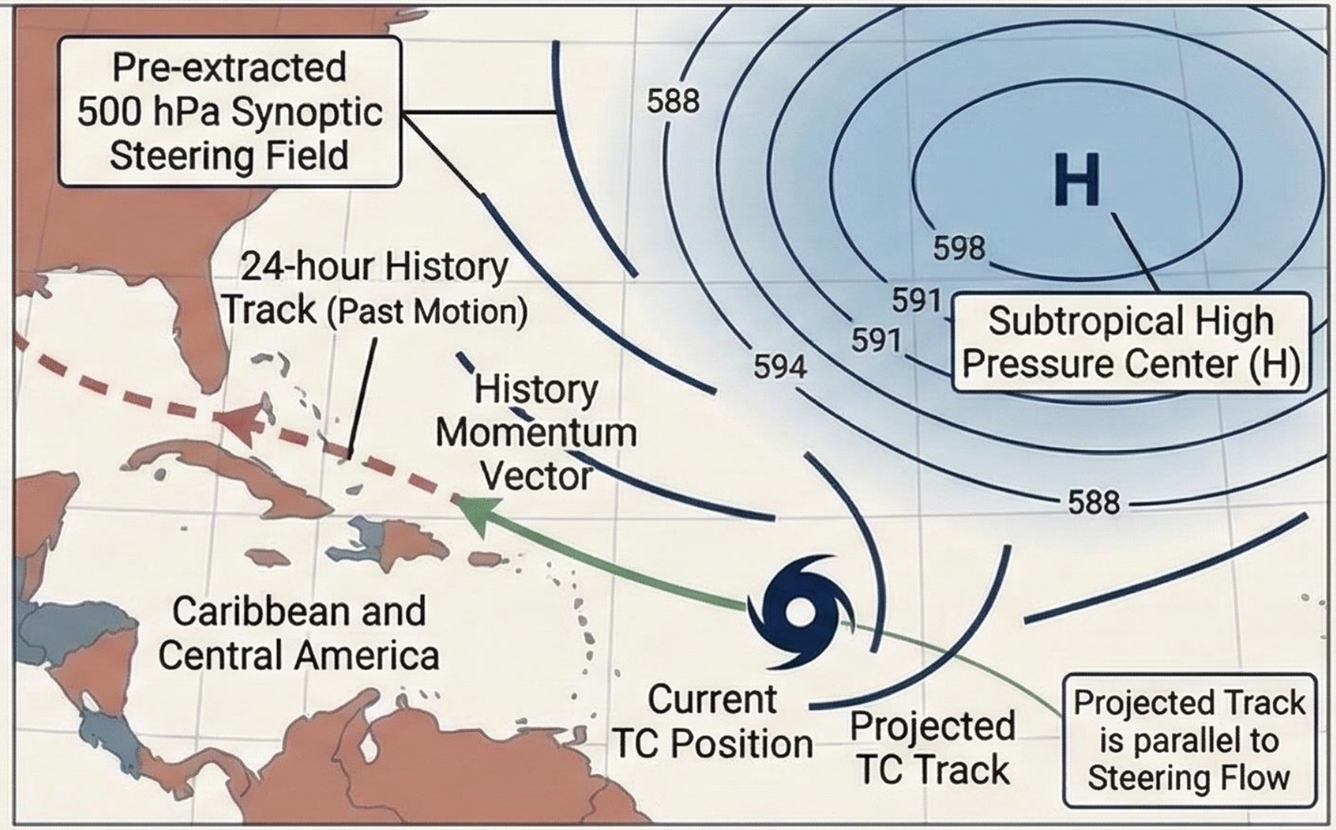
Data availability : Visualizations





The ‘Golden Branch’: Env-Data

- **Temporal Momentum:** 24-hour history of direction and intensity change.
- **Synoptic Steering:** The “Subtropical High Region” pre-extracted from 500 hPa Geopotential Height fields.
- **Context:** Movement velocity, month, and relative Earth location.



Mapping Data_3d to Physical Reality

To build effective models, tensors must be understood as physical phenomena.

200 hPa

Upper troposphere
Outflow layer (exhaust determining intensification).

500 hPa

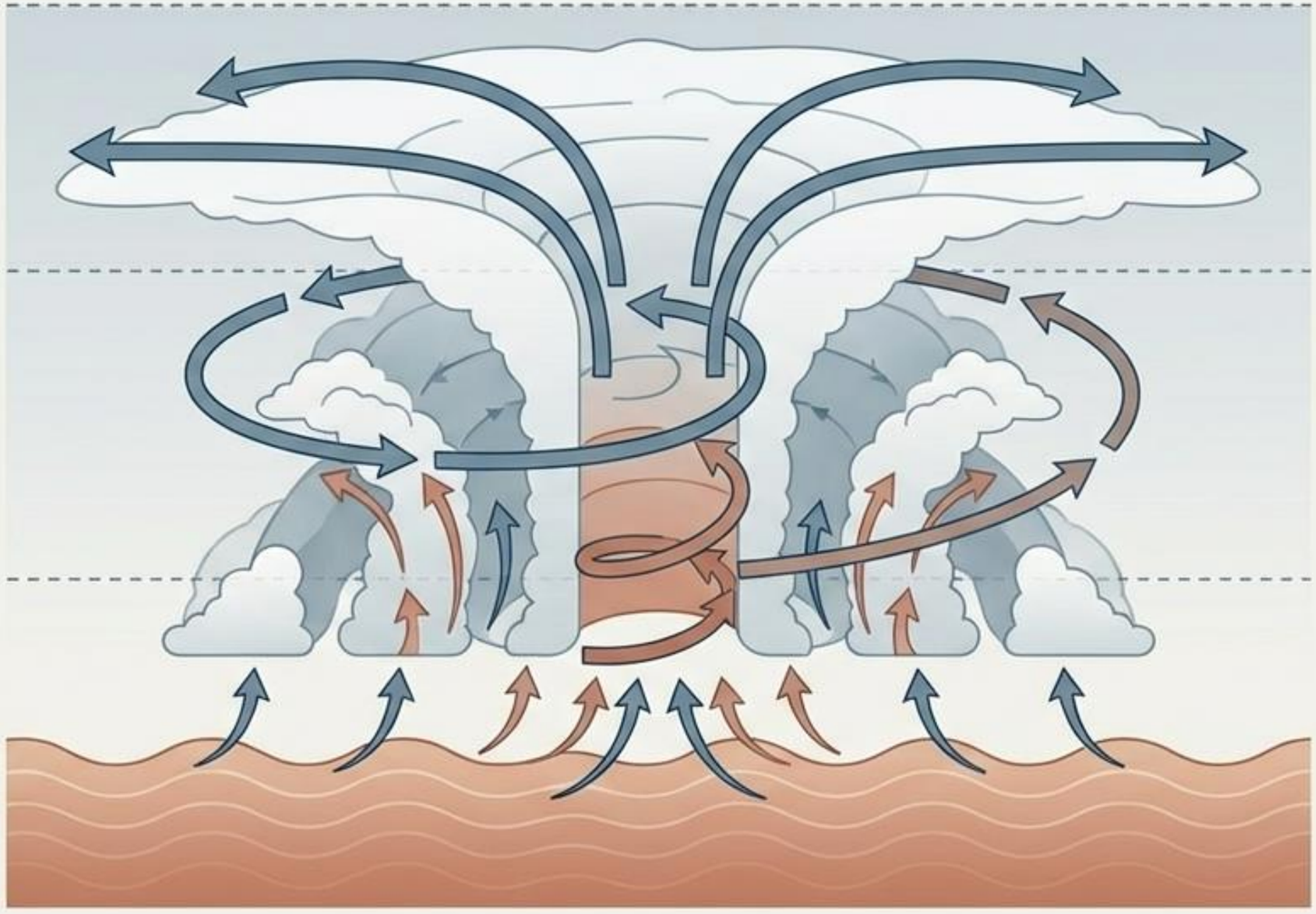
Mid-troposphere
Steering flow (the Subtropical High).

850 & 925 hPa

Boundary layer
Moisture inflow

SST

Sea Surface Temperature
Thermodynamic fuel



Optimizing Inherent Attributes for Deep Learning

Data_1d normalizes the cyclone's inherent tracking attributes to ensure compatibility with neural network activation functions.

Longitude Normalization

$$LONG_{norm} = \frac{LONG}{3600}$$

Latitude Normalization

$$LAT_{norm} = \frac{LAT+900}{1800}$$

Pressure Normalization

$$PRES_{norm} = \frac{PRES - 870}{1080}$$

Wind Speed Normalization

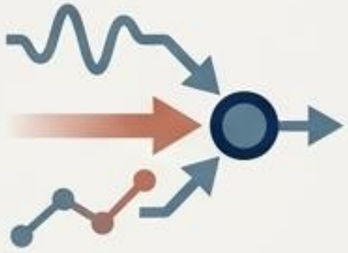
```
WND_norm = WND / 100
```

Research Frontiers for Group Projects

Four distinct pathways to leverage the TCND Test Subset for your final presentations.

1. Rapid Intensification

Quantify how anomalies in Sea Surface Temperature (SST) trigger 24-hour intensity spikes.



2. Multimodal Track Forecasting

Compare 1D coordinate models against models fused with 500 hPa Env-Data steering flows.

3. Basin Generalization

Train on the Western North Pacific; evaluate zero-shot performance on South Pacific cyclones.

4. Landfall Impact

Predict specific landfall timing and intensity decay using high-resolution 925 hPa boundary layers.

Technical Implementation & Quick Start

```
Dataset & Scripts:
github.com/xiaochengfuhuo/TropiCycloneNet-Dataset

Ingestion Pipeline:
data_1d = pd.read_csv('TCND_subset/Data_1d/...')
ds_3d = xr.open_dataset('TCND_subset/Data_3d/...')

Visualization:
# Run read_TCND.py to generate baseline plots
```

Scan to Access GitHub Repository

Reference

[1] https://www.un.org/en/climatechange/reports

[2] https://www.ncei.noaa.gov/access/billions/

[3] https://github.com/Zjut-MultimediaPlus/Benchmark-Dataset-and-Deep-Learning-Method-for-Global-Tropical-Cyclone-Forecasting

[4] https://github.com/xiaochengfuhuo/TropiCycloneNet-Dataset

[5] Hsiang, Solomon M., et al. "The Causal Effect of Environmental Catastrophe on Long-Run Economic Growth." National Bureau of Economic Research (2026). https://www.nber.org/system/files/working_papers/w20352/w20352.pdf

[6] Smith, A., et al. "The social costs of tropical cyclones." PMC - NIH (2026). https://pmc.ncbi.nlm.nih.gov/articles/PMC10667268/

[7] Chen, R., et al. "Avoidable Economic Losses from Influential Tropical Cyclones." MDPI Sustainability 18.4 (2026): 1845. https://www.mdpi.com/2071-1050/18/4/1845

[8] Chen, J., et al. "Benchmark dataset and deep learning method for global tropical cyclone forecasting." PubMed (2026). https://pubmed.ncbi.nlm.nih.gov/40595595/

[9] Huang, L., et al. "FuXi-TC: A generative framework integrating deep learning for tropical cyclone forecasting." arXiv (2026). https://arxiv.org/html/2508.16168v1

[10] Wang, Y., et al. "Accurate tropical cyclone intensity forecasts using a non-iterative spatiotemporal transformer model." ResearchGate (2026). https://www.researchgate.net/publication/398411677

[11] Liu, K., et al. "TropiCycloneNet framework: It includes TropiCycloneNet Dataset (TCND) and TropiCycloneNet model." ResearchGate (2026). https://www.researchgate.net/figure/TropiCycloneNet-framework-It-includes-TropiCycloneNet-Dataset-TCND-and-TropiCycloneNet_fig2_393267913

[12] Zhang, X., et al. "Benchmark dataset and deep learning method for global tropical cyclone forecasting." PMC - NIH (2026). https://pmc.ncbi.nlm.nih.gov/articles/PMC12219760/